

Aeration shall be automatically regulated by air control valve with electric actuator as per Section 7.8.18 installed for each aeration tank, based on measurement of the oxygen concentration in the individual tanks. If simultaneous denitrification shall be proposed, each tank shall be equipped with two oxygen measurement devices and transmitters (one oxygen probe in anoxic zone and one in aerobic zone). The output of the blowers for aeration shall be automatically controlled in correspondence with the actual oxygen concentration in the activated sludge tanks.

Flow meters (pitot tube or other reliable type) should be installed on each air pipes to aeration tanks individually.

The tanks shall be fitted with service bridges and shall be designed for easy emptying. Width of each service bridge shall be minimum 120 cm excluding the air pipes and balusters. Air pipe network on the aeration tanks shall be so designed that easy access to whole instrumentation shall not be hindered. Each tank can be disabled for individual maintenance if division is required. The drainage pit shall be designed and constructed on the bottom of each aeration tank. This drainage pit shall be equipped with the drainage valve that is extended spindle and manual operated.

Minimum number of activated sludge tanks for Stage 2	: 2 tanks
Minimum number of oxygen meters for each tank	: 2 units (1 for oxic and 1 for anoxic zone)
Minimum number of ORP meters for each tank	: 2 units (for anoxic zone)
Minimum number of NO ₃ analyzers	: 1 unit
Maximum dimensioning MLSS concentration	: 4000 mg/l

Safety Factor (SF) to be used for determining (aerobic) dimensioning sludge age shall be at least 1,45.

The sprinkler system shall be designed and installed along the service bridges to prevent and/or remove the surface scum in the aeration tanks. The treated water should be used for this purposes.

3.3.4.13 Methanol storage and dosing system

The methanol will be transported to the site by tanker trucks. Methanol shall be stored in outdoor, above ground tanks with floating roof at atmospheric pressure, with aboveground piping in pipe racks. The storage tanks shall be roofed with metal shades made from galvanized steel, aluminum or equivalent material. The tanks shall be grounded to avoid hazards associated with static discharge. Ignition control shall be by flame arresters. All piping and valves subject to carrying methanol shall be consistently labelled and direction of flow shall be indicated. All storage materials shall require berming and adequate ventilation. Berming shall be with concrete.

In the design and construction of methanol facilities, an extra precaution shall be made for leak detection, toxic hazard and availability of alcohol-compatible fire suppression foam regarding Section 2.8 and 2.9.

Electrical equipment within the proximity of methanol storage and handling shall be explosion proof. Grounding is especially important in protecting methanol from accidental ignition resulting from static discharge. The grounding straps shall be equipped with carbide-tipped clamps to ensure electrical contact through nonconductive surface coatings, such as paint. Tanks shall be fitted with dip-tube-